

Blending

Problem

- Find the optimal purchasing and sales policy.

Sets:

- Oils: $o = 1(V1), 2(V2), 3(N1), 4(N2), 5(N3)$.

Parameters

- Sale price of final product: $SellPrice$
- Purchase price of oil o : $Cost_o$
- Hardness of oil o : $Hardness_o$
- Lower hardness bound: $LowerHardness$
- Upper hardness bound: $UpperHardness$
- Refinery capacity of Vegetable oil:
 $RefCapVeg$
- Refinery capacity of Non-Vegetable oil:
 $RefCapNonVeg$

Decision variables

- Amount of oil o bought and used: x_o

Model

Objective:

- Objective is profit to be maximized:

$$\sum_o SellPrice \cdot x_o - \sum_o Cost_o \cdot x_o$$

Constraints:

- Lower hardness:

$$\sum_o Hardness_o \cdot x_o \geq LowerHardness \cdot \sum_o x_o$$

- Upper hardness:

$$\sum_o Hardness_o \cdot x_o \leq UpperHardness \cdot \sum_o x_o$$

- VO refinery capacity:

$$\sum_{o=1}^2 x_o \leq RefCapVeg$$

- NO refinery capacity:

$$\sum_{o=3}^5 x_o \leq RefCapNonVeg$$

- Positive buying:

$$x_o \geq 0, \forall o$$

The full model in Julia/JuMP, available with the name

Blending.jl

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#####
# Blending Assignment, "Mathematical Programming Modelling" (42112)

#####
# Intro definitions
using JuMP
using HiGHS
#####

#####

# INDEXES
Oils = ["VEG1" "VEG2" "OIL1" "OIL2" "OIL3"]
O = length(Oils)

# DATA
```

```

Cost      = [110 120 130 110 115] # cost of buing the different oils
Hardness = [8.8 6.1 2.0 4.2 5.0] # hardness of the different oils
SellPrice = 150 # selling price for final product
RefCapVeg = 200
RefCapNonVeg = 250
LowerHardness = 3
UpperHardness = 6
*****

#*****
# Model
blending = Model(HiGHS.Optimizer)

# how much buy of different oils
@variable(blending, x[1:0] >= 0)

# Maximize profit
@objective(blending, Max,
            sum(SellPrice*x[o] for o=1:0) -
            sum(Cost[o]*x[o] for o=1:0))

# lower hardness
@constraint(blending,
            sum(Hardness[o]*x[o] for o=1:0) >=
            LowerHardness*sum(x[o] for o=1:0))

# upper hardness
@constraint(blending,
            sum(Hardness[o]*x[o] for o=1:0) <=
            UpperHardness*sum(x[o] for o=1:0))

# veg capacity
@constraint(blending,
            sum(x[o] for o=1:2) <= RefCapVeg)

# non-veg capacity
@constraint(blending,
            sum(x[o] for o=3:0) <= RefCapNonVeg)

print(blending)
*****

#*****
# solve

```

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optimize!(blending)
println("Termination status: $(termination_status(blending))")
#####

#####
println("-----");
if termination_status(blending) == MOI.OPTIMAL
    println("RESULTS:")
    println("Blending result: $(objective_value(blending))")
    for o = 1:O
        println(" bought oil: $(value(x[o]))")
    end
else
    println(" No solution")
end
println("-----");
#####

#####
println("Successfull end of $(PROGRAM_FILE)")
#####

```

Comments to the Julia/JuMP program:

- Notice: The mixing constraints, lower hardness and upper hardness, were easy to linearise by multiplication.